

A Counterexample to Conjecture 00000000013

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Abstract

Conjecture 00000000013 asserts that for every $n \geq 3$ the interval $[2^n, 2^{n+1})$ contains a prime with exactly three binary ones. It fails at $n = 8$: all twenty-eight such integers are composite. The refutation is formalized in Lean 4 against Mathlib, with the conjecture itself stated in the formalization.

1 Statement

Conjecture 00000000013 claims that for every integer $n \geq 3$ there is a prime p with

$$2^n \leq p < 2^{n+1}$$

whose binary expansion contains exactly three 1's. We disprove it by exhibiting $n = 8$.

2 The counterexample

Theorem 1. *No prime p with $2^8 \leq p < 2^9$ has exactly three 1's in its binary expansion.*

Proof. An integer of $[2^8, 2^9)$ has its top bit at position 8, so it has exactly three binary ones precisely when it is of the form

$$2^8 + 2^a + 2^b, \quad 0 \leq b < a < 8.$$

There are $\binom{8}{2} = 28$ such integers. Each is composite, as witnessed by the proper divisor listed beside it:

N	div	N	div	N	div	N	div
259	7	261	3	262	2	265	5
266	2	268	2	273	3	274	2
276	2	280	2	289	17	290	2
292	2	296	2	304	2	321	3
322	2	324	2	328	2	336	2
352	2	385	5	386	2	388	2
392	2	400	2	416	2	448	2

Every entry in the second column is a divisor d with $1 < d < N$, so each N is composite. Twenty-one of the twenty-eight are even and exceed 2; the seven odd ones are

$$\begin{aligned} 259 &= 7 \cdot 37, & 261 &= 3^2 \cdot 29, & 265 &= 5 \cdot 53, & 273 &= 3 \cdot 7 \cdot 13, \\ 289 &= 17^2, & 321 &= 3 \cdot 107, & 385 &= 5 \cdot 7 \cdot 11. \end{aligned}$$

□

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Since the conjecture asserts the existence of such a prime for *every* $n \geq 3$, the single value $n = 8$ refutes it.

Remark 2. $n = 8$ is not isolated. A direct search finds that $n = 25$ and $n = 32$ also admit no such prime: of the $\binom{25}{2} = 300$ and $\binom{32}{2} = 496$ candidates in the respective intervals, none is prime. These were checked by computation and are not part of the Lean development, where a kernel-level decision procedure over intervals of that size is not practical.

3 Formal verification

The accompanying Lean 4 project formalizes the refutation against Mathlib.

- `ConjectureHolds` — the conjecture itself:

$$\forall n \geq 3, \exists p, 2^n \leq p \wedge p < 2^{n+1} \wedge (\text{Nat.bits } p).\text{count true} = 3 \wedge \text{Nat.Prime } p.$$

The digit condition counts the `true` entries of the binary expansion `Nat.bits p` supplied by Mathlib, rather than a parametrization assumed to describe it, and primality is Mathlib's `Nat.Prime`.

- `card_candidates` — the interval contains exactly 28 integers with three binary ones, matching $\binom{8}{2}$.
- `no_three_one_prime_at_eight` — no such integer is prime, decided by exhausting all 256 integers of $[2^8, 2^9)$.
- `conjecture_00000000013_false` — $\neg \text{ConjectureHolds}$, by instantiating the conjecture at $n = 8$.

Because the conjecture is stated in the formalization, the final theorem is a literal formalization of its negation rather than of facts that merely entail it.

The project builds with `lake build` on `leanprover/lean4:v4.33.1` with Mathlib `v4.33.1`. It contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms of its own: `#print axioms conjecture_00000000013_false` reports exactly the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.